

RQ Answers

omitted
2024-04-05

Data overview

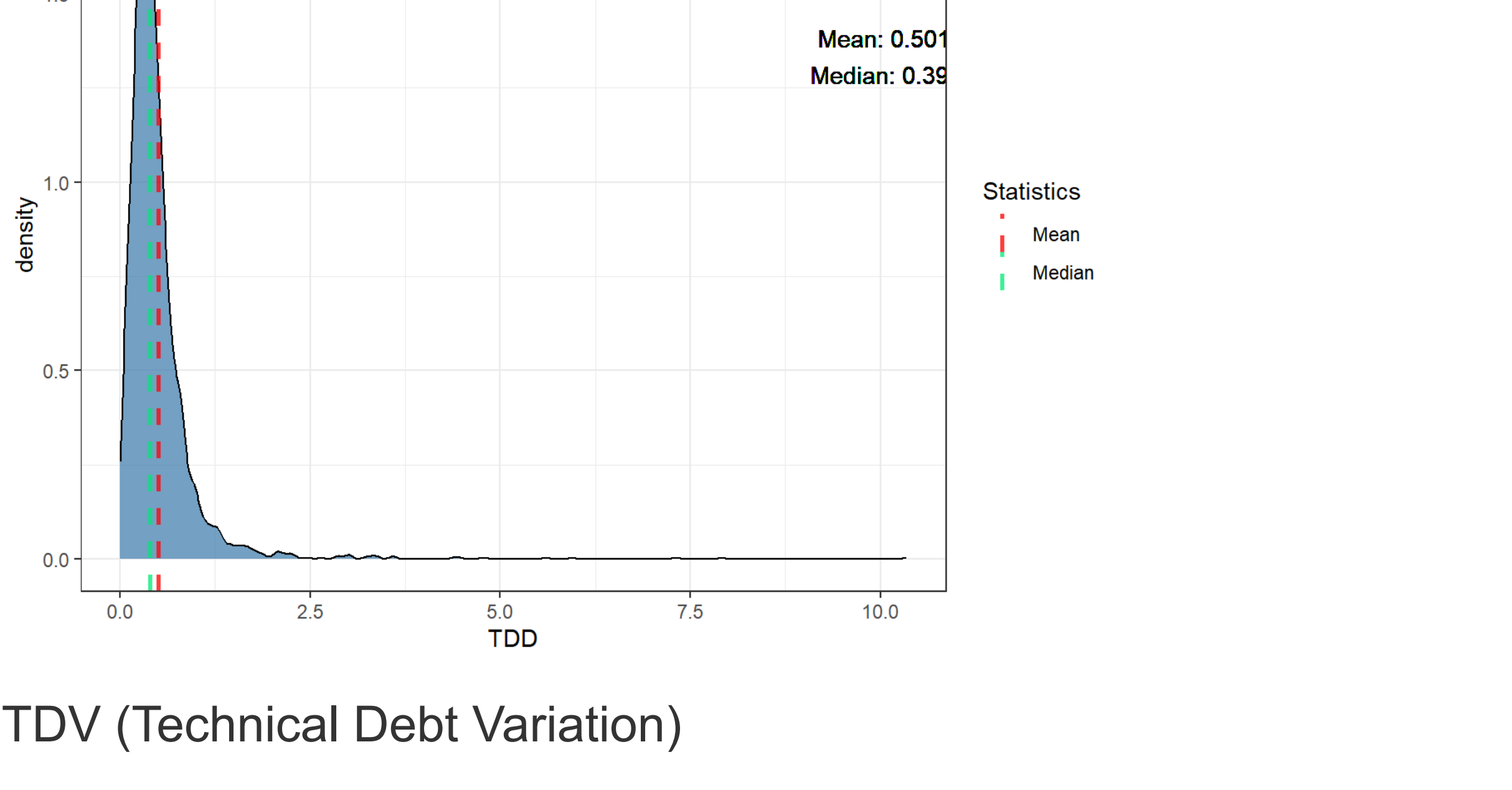
glimpse(all_data)	
## Rows: 247,365	
## Columns: 11	
## \$ repo	<chr> "accumulo", "accumulo", "accumulo", "accumulo", "a...
## \$ pr_number	<int> 1302, 1302, 1302, 1302, 1302, 1302, 1302, 1302, 13...
## \$ rule	<chr> "java:S1192", "java:S3776", "java:S1319", "java:S1...
## \$ severity	<chr> "CRITICAL", "CRITICAL", "MINOR", "CRITICAL", "CRIT...
## \$ file	<chr> "core/src/main/java/org/apache/accumulo/core/metad...
## \$ type	<chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...
## \$ status	<chr> "OPEN", "OPEN", "OPEN", "OPEN", "CLOSED", "OPEN", ...
## \$ debt	<int> 8, 6, 10, 5, 6, 2, 5, 15, 30, 109, 30, 30, 5, 5, 5...
## \$ ncloc_affected_file	<int> 347, 347, 347, 347, 347, 347, 347, 347, 252, 252, ...
## \$ complexity	<int> 68, 68, 68, 68, 68, 68, 68, 68, 86, 86, 86, 86...
## \$ origin	<chr> "NEW", "NEW", "NEW", "NEW", "PRE-EXISTING", "PRE-E...

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 1959 rows.
## i Please consider using 'annotate()' or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 1959 rows.
## i Please consider using 'annotate()' or provide this layer with data containing
##   a single row.
```

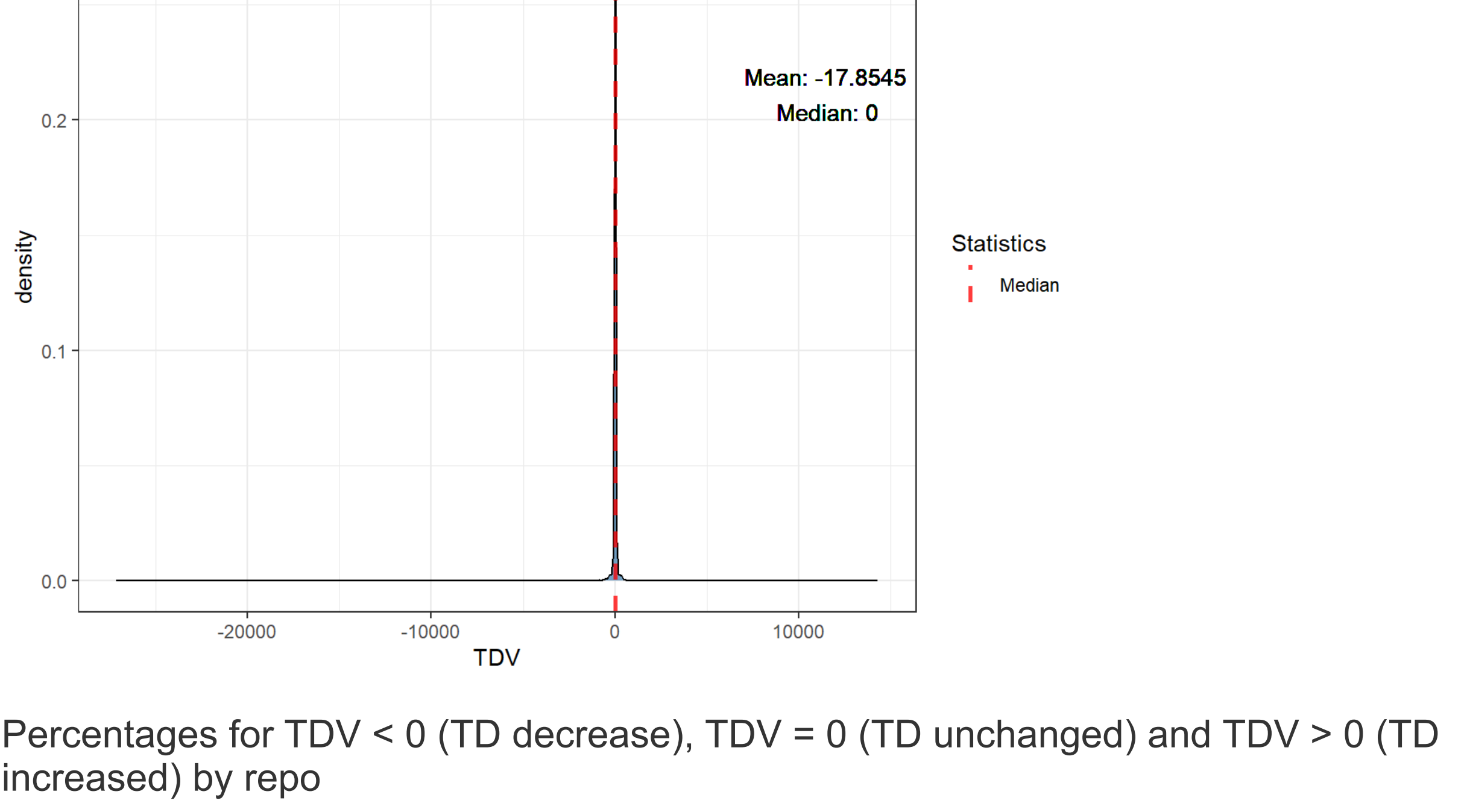


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 1959 rows.
## i Please consider using 'annotate()' or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 1959 rows.
## i Please consider using 'annotate()' or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 12 x 4		tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero
## repo	<chr>	<dbl>	<dbl>	<dbl>
## 1 accumulo		26.2	47.0	26.9
## 2 cayenne		12.7	56.4	30.9
## 3 commons-collections		35.7	53.6	10.7
## 4 commons-io		32.9	50.7	16.4
## 5 commons-lang		13.9	78.7	7.38
## 6 helix		19.2	37.0	43.8
## 7 httpcomponents-client		19.2	55.8	25
## 8 maven-surefire		24	68	8
## 9 opennlp		44.7	33.0	22.3
## 10 struts		37.1	36.4	26.4
## 11 wicket		10	57.5	32.5
## 12 zookeeper		17.6	52.9	29.4

Mean of percentages of TDV (mean of repo percentages)

Percentages of pre-existing TDV by repo

A tibble: 12 x 3

Percentages of pre-existing TDV by repo

## # A tibble: 12 x 3		preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero
## repo	<chr>	<dbl>	<dbl>
## 1 accumulo		57.6	42.4
## 2 cayenne		75.9	24.1
## 3 commons-collections		53.8	46.2
## 4 commons-io		62.5	37.5
## 5 commons-lang		81.2	18.8
## 6 helix		56.5	43.5
## 7 httpcomponents-client		69.5	30.5
## 8 maven-surefire		68	32
## 9 opennlp		44.0	56.0
## 10 struts		51.8	48.2
## 11 wicket		64.1	35.9
## 12 zookeeper		67.6	32.4

```
## # i abbreviated name: 'preexisting_tdv_greater_than_zero'
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

##	<chr>	<dbl>	<dbl>
##	1 accumulo	75.3	24.7
##	2 cayenne	90.9	9.09
##	3 commons-collections	66.7	33.3
##	4 commons-io	84	16

Percentages for new TDV by repo

## # A tibble: 12 x 3		new_tdv_equal_to_zero	new_tdv_greater_than_zero
## repo	<chr>	<dbl>	<dbl>
## 1 accumulo		75.3	24.7
## 2 cayenne		90.9	9.09
## 3 commons-collections		66.7	33.3
## 4 commons-io		84	16
## 5 commons-lang		85	15
## 6 helix		75.9	24.1
## 7 httpcomponents-client		94.7	5.26
## 8 maven-surefire		100	0
## 9 opennlp		95.8	4.17
## 10 struts		91.4	8.57
## 11 wicket		94.1	5.88
## 12 zookeeper		78.6	21.4

Mean of percentages of new TDV (mean of repo percentages)

RQZ

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_j$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 x 13		rank	accumulo	cayenne	'commons-collections'	'commons-io'	'commons-lang'
##	rank	<int>	<chr>	<chr>	<chr>	<chr>	<chr>
## 1	1	java:S1117	---	java:S...	java:S112	- 10	java:S4719 ... java:S4165
## 2	2	java:S1112	---	java:S...	java:S1125	- 8	java:S1192 ... java:S1192
## 3	3	java:S1192	---	java:S...	java:S1170	- 6	java:S4842 ... java:S125
## 4	4	java:S2589	---	java:S...	java:S1197	- 5	java:S899 ... java:S2293
## 5	5	java:S1125	---	java:S...	java:S1874	- 5	java:S1125 ... java:S3878
## 6	6	java:S1125	---	java:S...	java:S3008	- 5	java:S1874 ... java:S1612
## 7	7	java:S3776	---	java:S...	java:S1192	- 4	java:S2129 ... java:S100
## 8	8	java:S2293	---	java:S...	java:S125	- 4	java:S2293 ... java:S1611
## 9	9	java:S1155	---	java:S...	java:S1134	- 4	java:S1481 ... java:S2129
## 10	10	java:S1135	---	java:S...	java:S1126	- 2	java:S1612 ... java:S3984

```
## # i 7 more variables: helix <chr>, 'httpcomponents-client' <chr>,
## # 'maven-surefire' <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## # zookeeper <chr>
```

TOP 10 fixed

TOP 10 fixed rules by *PositionScore* metric.

## # A tibble: 57 x 2		rule	position_score
##	<chr>	<chr>	<dbl>
## 1	java:S1192		41
## 2	java:S1874		84
## 3	java:S112		90
## 4	java:S2293		94
## 5	java:S3776		102
## 6	java:S100		103
## 7	java:S1161		107
## 8	java:S3740		108
## 9	java:S1125		111
## 10	java:S117		113

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 x 13		rank	accumulo	cayenne	'commons-collections'	'commons-io'	'commons-lang'
##	rank	<int>	<chr>	<chr>	<chr>	<chr>	<chr>
## 1	1	java:S117	---	java:S...	java:S1192	- 336	java:S1192 ... java:S1192
## 2	2	java:S2589	---	java:S...	java:S125	- 215	java:S100 ... java:S100
## 3	3	java:S1192	---	java:S...	java:S108	- 174	java:S899 ... java:S1168
## 4	4	java:S101	---	java:S...	java:S1117	- 122	java:S4842 ... java:S1133
## 5	5	java:S1125	---	java:S...	java:S2293	- 93	java:S4719 ... java:S3776
## 6	6	java:S1125	---	java:S...	java:S1125	- 89	java:S1874 ... java:S4165
## 7	7	java:S3776	---	java:S...	java:S2160	- 57	java:S117 ... java:S125
## 8	8	java:S2293	---	java:S...	java:S128	- 52	java:S1133 ... java:S1172
## 9	9	java:S116	---	java:S...	java:S3776	- 52	java:S108 ... java:S1149
## 10	10	java:S2142	---	java:S...	java:S2065	- 51	java:S3878 ... java:S3878

```
## # i 7 more variables: helix <chr>, 'httpcomponents-client' <chr>,
## # 'maven-surefire' <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## # zookeeper <chr>
```

TOP 10 unfixed

TOP 10 unfixed rules by *PositionScore* metric.

## # A tibble: 50 x 2		rule	position_score
##	<chr>	<chr>	<dbl>
## 1	java:S1192		22
## 2	java:S3776		85
## 3	java:S100		100
## 4	java:S106		100
## 5	java:S117		101
## 6	java:S1133		103
## 7	java:S1874		107
## 8	java:S3740		108
## 9	java:S1135		111
## 10	java:S1125		112

Outliers

Higher TDD

## # A tibble: 1,959 x 5		repo	pr_number	total_debt	total_ncloc	tdt
##	<chr>	<int>	<chr>	<int>	<int>	<dbl>
## 1	wicket	676		600	58	10.3
## 2	helix	1975		261	33	7.91
## 3	wicket	708		3157	432	7.31
## 4	accumulo	2922		410	69	5.94
## 5	accumulo	4215		140	25	5.6
## 6	wicket	709		2265	474	4.78
## 7	opennlp	563		1035	232	4.46
## 8	opennlp	561		74742	17056	4.38
## 9	wicket	351		2159	599	3.60
## 10	opennlp	569		1161	325	3.57

Higher TDV

## # A tibble: 1,959 x 3		pr_number	repo	tdv
##	<int>	<chr>	<int>	<dbl>
## 1	2022	accumulo		14300
## 2	1946	accumulo		3075
## 3	2583	accumulo		2078
## 4	2335	accumulo		1492
## 5	1891	accumulo		1286
## 6	1605	accumulo		1082
## 7	4054	accumulo		593
## 8	427	struts		579
## 9	483	httpcomponents-client		525
## 10	559	struts		510

Lower TDV

## # A tibble: 1,959 x 3		pr_number	repo	tdv
##	<int>	<chr>	<int>	<dbl>
## 1	799	struts		-27133
## 2	561	opennlp		-12880
## 3	3604	accumulo		-6508
## 4	3402	accumulo		-5186
## 5	670	struts		-1655
## 6	3117	accumulo		-1262
## 7	3229	accumulo		-899
## 8	484	httpcomponents-client		-884
## 9	431	opennlp		-813
## 10	2705	accumulo		-724

Extra

Number of PRs with pre-existing TD

## n	
## 1	18504

Number of java:S1192 issues that affect test code

## n	
## 1	18346

Number of java:S1192 issues that affect not test code

## n	
## 1	8346

Summary NCLOC

## ncloc	
## Min.	3.0
## 1st Qu.	256.5
## Median	651.0
## Mean	2327.5
## 3rd Qu.	1772.5
## Max.	240475.0